THE FIRST IT COLLEGE OF NEPAL

IN ACADEMIC COLLABORATION WITH ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION

Design and Development of an AI-Powered Cloud-Based Public Notice Management System for Nepal

Ashok Bhattarai | NP069811 | NP3F2509IT

B.Sc. (Hons) in Information Technology (NP3F2509IT)

Supervisor: **Dr. Laxman Mandal** | 2nd Marker: **Manish Kumar Tamang**

Live system: suchanaai.tech

SCAN TO RATE / VOTE

Introduction & Aim

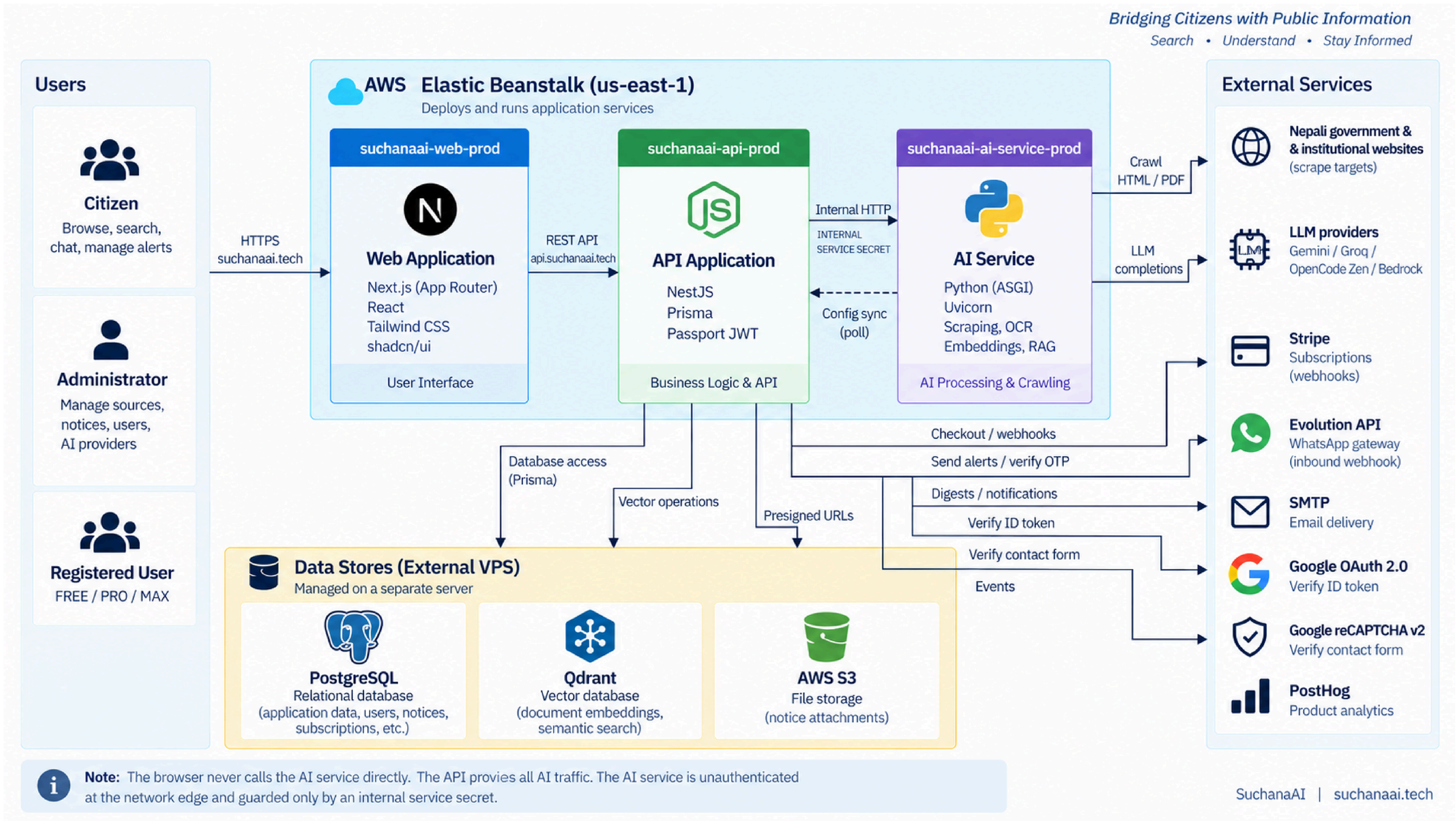
SuchanaAI is a cloud-based platform that consolidates public notices from Nepalese government and institutional portals into one responsive interface. It combines automated ingestion, structured search, AI summaries and document-grounded question answering.

Aim: automate the collection, categorisation, summarisation and intelligent delivery of public notices so citizens can find and understand official information more easily.

Problem Statement

- Official notices are fragmented across many independent portals with inconsistent structures and update patterns.
- Scanned PDFs and image-based notices are often not machine-searchable, creating extra effort and accessibility barriers.
- Citizens repeatedly visit multiple sites to track vacancies, exams, tenders and policy updates.
- There is no single interface combining aggregation, search, alerts, AI summaries and document-based Q&A.;

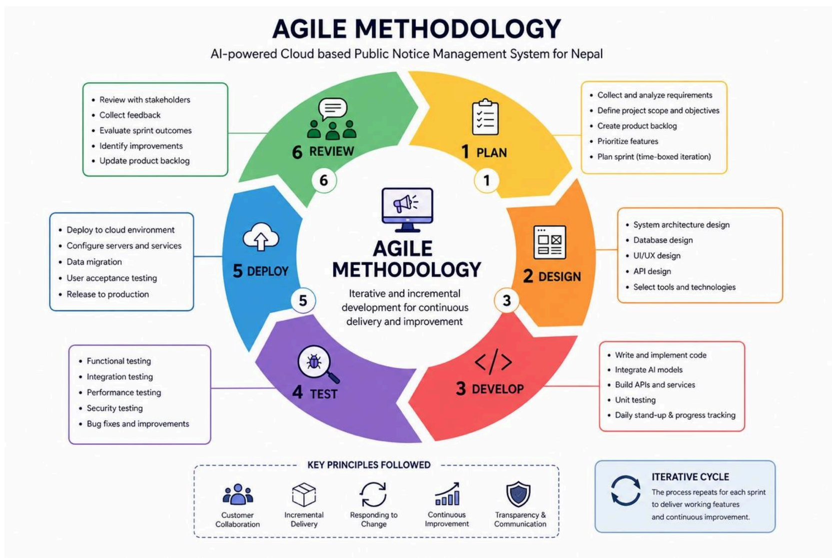
System Architecture



Objectives

- 1. Analyse current publication practices, extraction challenges and citizen access barriers.
- 2. Build automated collection, extraction, normalisation and storage for multiple Nepalese notice sources.
- 3. Integrate AI classification, plain-language summaries and retrieval-augmented question answering.
- 4. Develop, deploy and evaluate a cloud-hosted web application with unified search, filters, browsing and alerts.

Methodology - Agile



Iterative cycle: Plan → Design → Develop → Test → Deploy → Review.

The Agile approach supports continuous feedback across scraping, AI, backend and frontend work. Testing is performed throughout each increment so changes can be refined without restarting the full development process.

SDG Goal 9

Industry, Innovation and Infrastructure

A stronger digital foundation for access to public information

Project alignment

SuchanaAI applies cloud infrastructure, AI-assisted information processing and a scalable digital platform to improve how Nepalese public notices are collected, organised and accessed.

Measurable contribution

- Digital public-information infrastructure
- Innovation in notice discovery and understanding
- More efficient, inclusive access to civic information

INFRASTRUCTURE
Cloud-hosted, scalable public platform

INNOVATION
AI summaries, OCR and grounded RAG

System / Product Features

- Unified notice aggregation and category/source/date filtering
- AI-generated plain-language summaries
- Natural-language RAG search with grounded source results
- Saved notices, alert rules and user dashboards
- Admin source health, scraping control and notice management
- Role-based authentication plus email / WhatsApp notification channels

Technology Stack

FRONTEND
Next.js - React - Tailwind CSS - shadcn/ui

BACKEND & API
NestJS - Prisma - REST - Passport/JWT

AI & INGESTION
Python ASGI - Uvicorn - Crawl4AI - OCR - Embeddings - RAG

DATA & CLOUD
PostgreSQL - Qdrant - AWS Elastic Beanstalk - AWS S3

INTEGRATIONS
Google OAuth - reCAPTCHA - SMTP - WhatsApp API - Stripe - PostHog

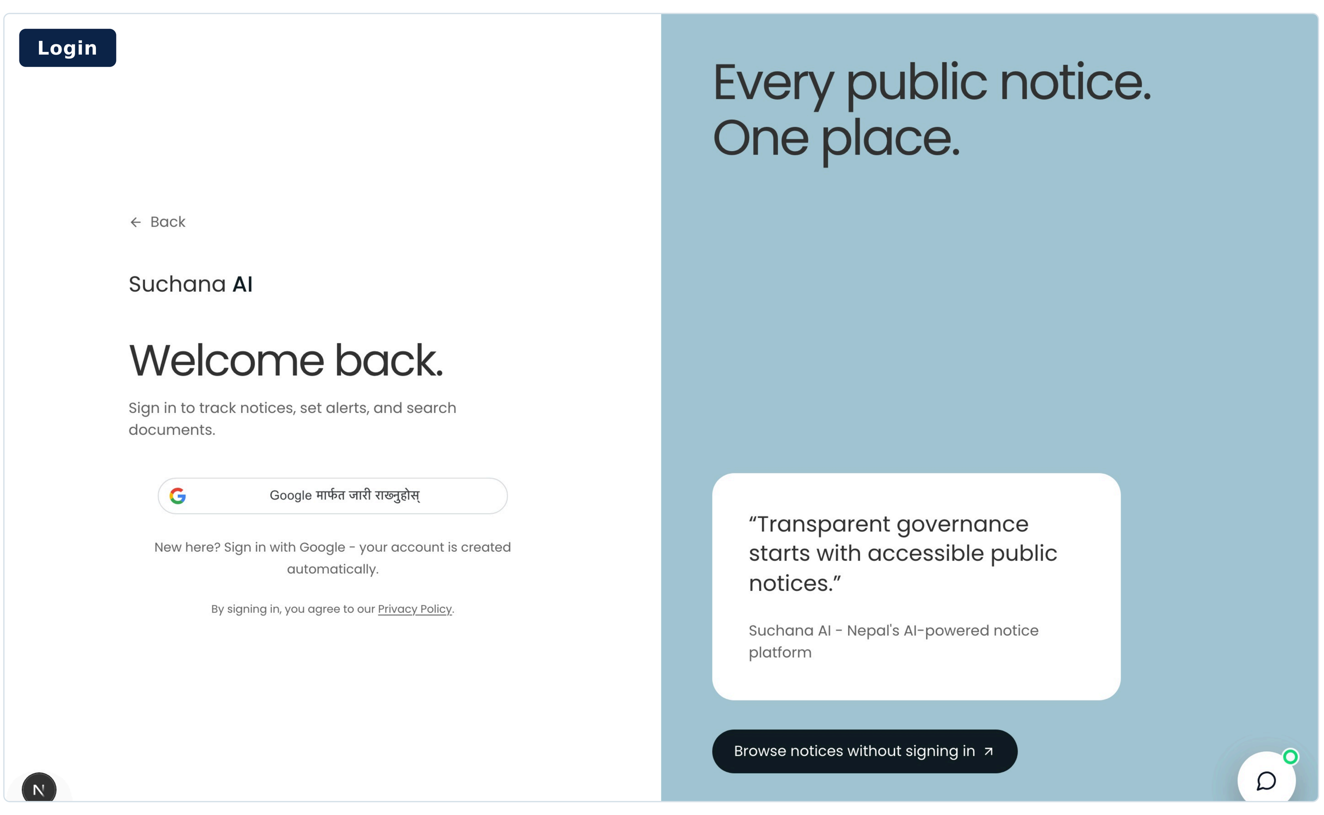
SECURITY
HTTPS - JWT - OAuth 2.0 - reCAPTCHA - environment secrets

OPERATIONS
Health checks - scheduled scraping - analytics - rollback verification

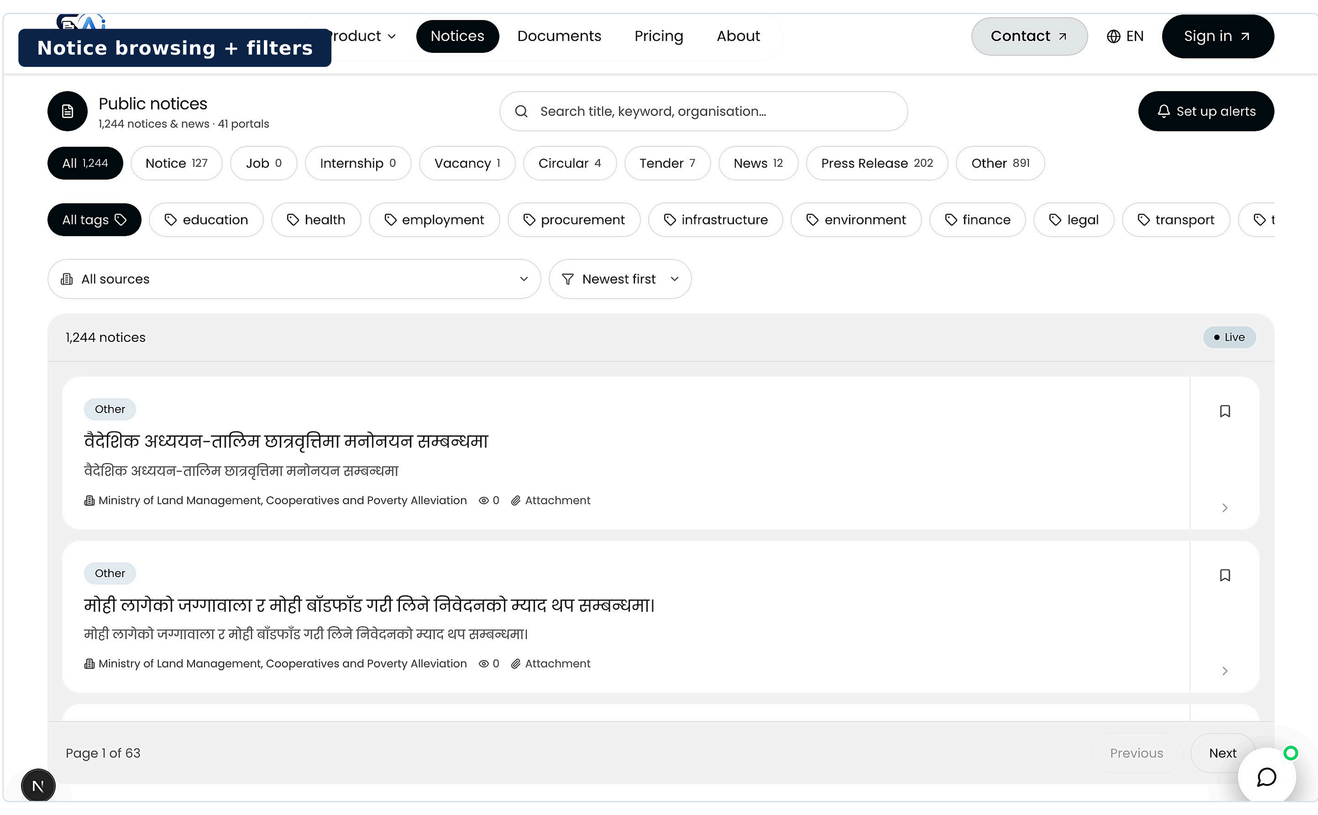
DEPLOYMENT
Independent web, API and AI services with external data stores

System Output - Implemented Interface

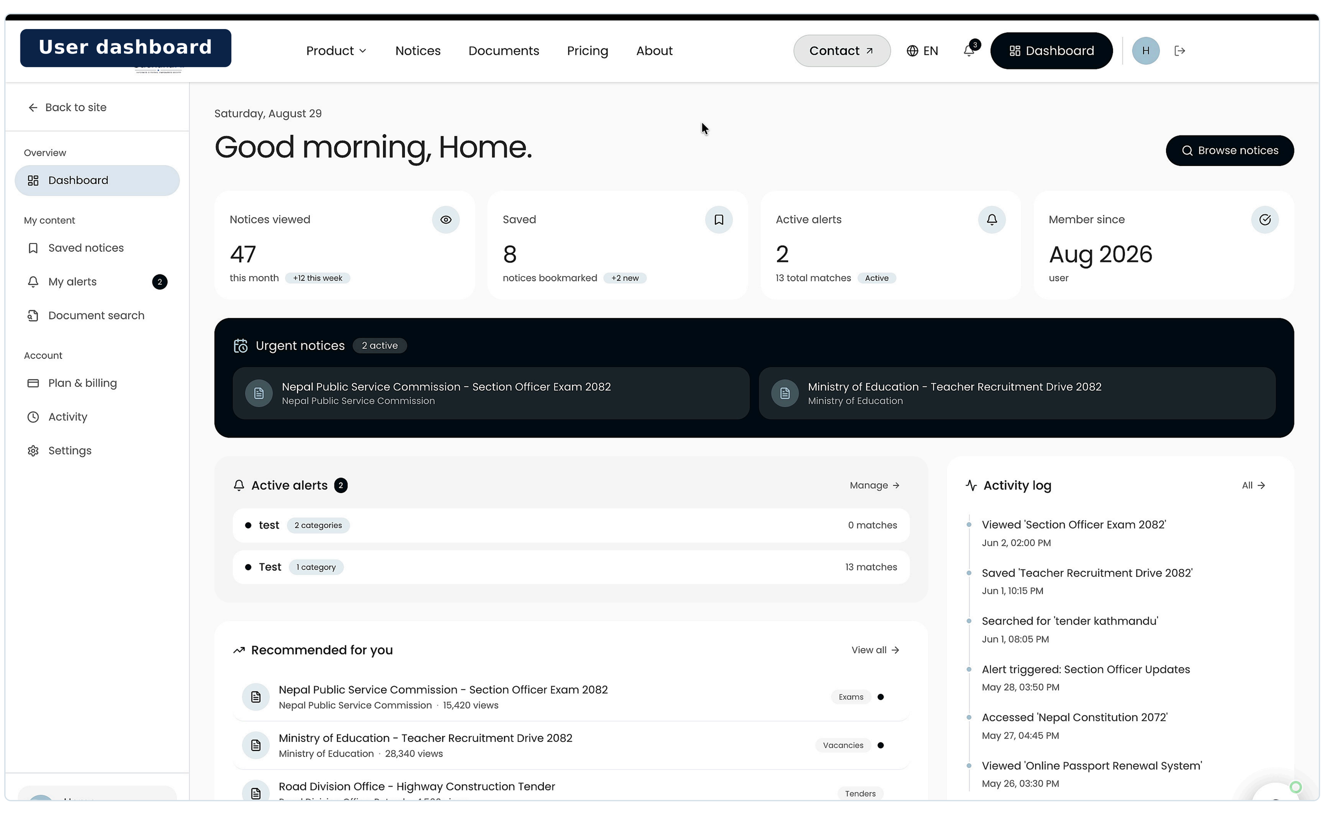
Login



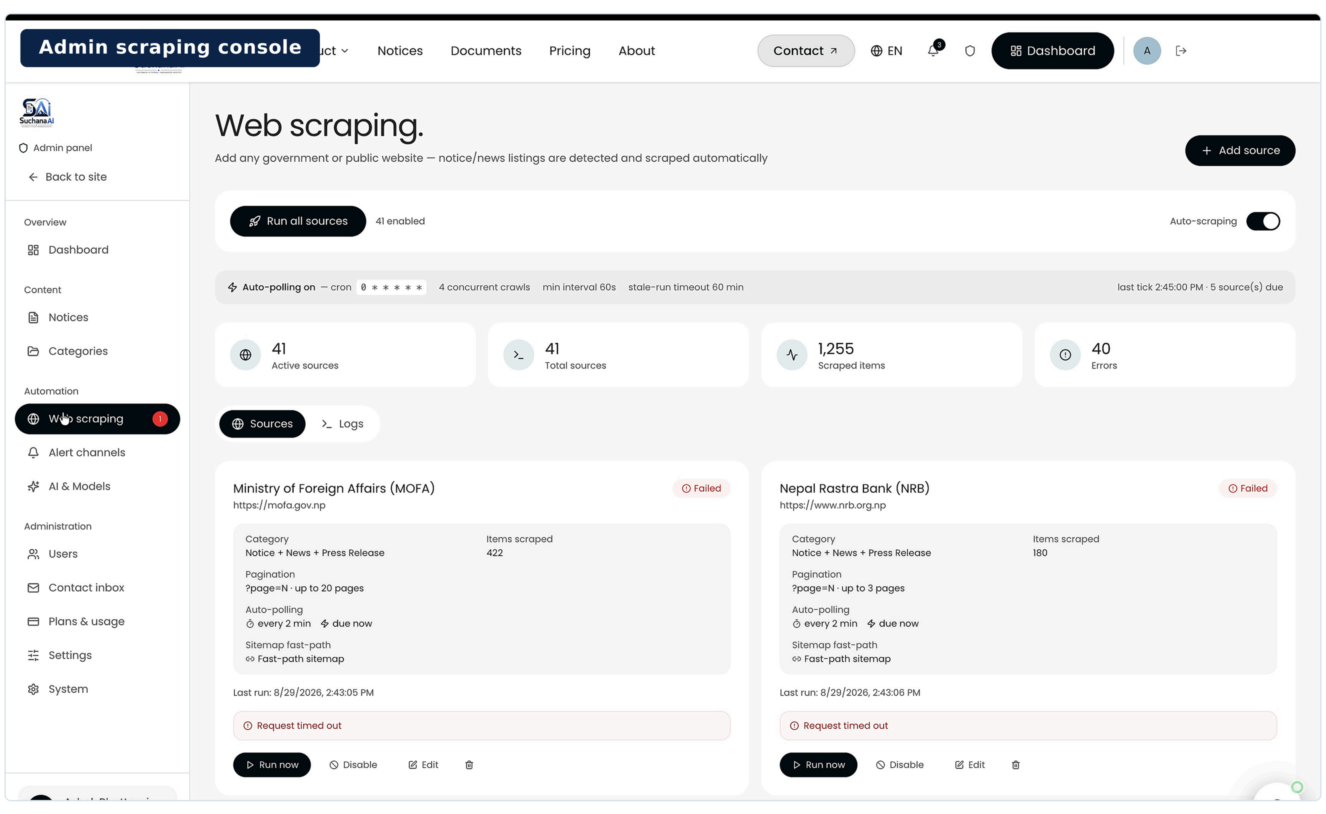
Notice browsing + filters



User dashboard



Admin scraping console



Testing & Evaluation

16	5	8	<120 ms	<250 ms
Unit tests passed	UAT participants	UAT scenarios	p95 notices-list response	Top-5 vector retrieval

Functional verification

- Registration, authentication, browse, filter and save workflows passed.
- Scraping parsing, malformed-source handling and record access were verified.
- RAG retrieval returned document-grounded results with controlled access.

User & operational validation

- Five target users completed eight scripted scenarios covering core journeys.
- Feedback improved filter wording, guidance and empty-state presentation.
- Responsive rendering, health checks, schedules and rollback behaviour passed.

Coverage matrix - final verification

PASS Registration	PASS Login / Logout	PASS Browse / Filter	PASS Save Notice
PASS Scraping Parse	PASS RAG Retrieval	PASS Alert Rules	PASS Record Access

Result: All planned functional, user, performance and operational checks passed.

Conclusion

SuchanaAI delivers a coherent cloud platform that links notice ingestion, AI-assisted understanding, search, RAG and administrative control in one system. The implemented modules are traceable to design artefacts and test evidence, with stable retrieval and usable end-to-end workflows.

Outcomes: a unified searchable notice repository, AI-assisted understanding, grounded RAG answers, user alerts and an administrator control surface.

Key limitations: scraper selectors require maintenance when source portals change, and OCR quality can limit retrieval from scanned documents containing mixed Nepali and English text.

Project Impact & Future Scope

SuchanaAI strengthens access to public information through unified notice aggregation, intelligent search and document-grounded AI.

Expected impact

- Reduces repeated visits to fragmented public-sector portals.
- Improves discovery through categories, filters and semantic search.
- Supports understanding with summaries and grounded RAG answers.
- Enables timely action through saved notices and alerts.

Future scope

- Expand national, provincial and local-government source coverage.
- Improve mixed Nepali-English OCR and summary evaluation.
- Add mobile-first delivery and broader accessibility support.
- Strengthen selector repair, monitoring, privacy and reliability.

References

Janssen, M., Charalabidis, Y., & Zuiderwijk, A. (2012). Benefits, adoption barriers and myths of open data and open government. Information Systems Management, 29(4), 258-268. https://doi.org/10.1080/10580530.2012.716740

Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2018). Artificial intelligence and the public sector: Applications and challenges. International Journal of Public Administration, 42(7), 67-79. https://doi.org/10.1080/01900692.2018.1498103

Varade, S., Sayyed, E., Nagtode, V., & Shinde, S. (2021). Text summarization using extractive and abstractive methods. ITM Web of Conferences, 5(3), 78-89. https://doi.org/10.1051/itmconf/2021400302384

Bale, A. S., Ghorpade, N., Kamalesh, S., & S, R. (2022). Web scraping approaches and their performance on modern websites. Third International Conference on Electronics and Sustainable Communication Systems, 8(1), 25-56. https://doi.org/10.1109/ICESC54411.2022.9885689

Shakya, S., Basnet, A., Sharma, S., & Gurung, A. (2023). Optical character recognition for Nepali, English character and simple sketch using neural network. Recent Advances in Information and Communication Technology, 6(1), 45-53. https://doi.org/10.1007/978-3-319-40415-8_6

Sharma, B. P. (2025). Use of information and communication technology (ICT) and e-governance in Dhangadhi Sub-Metropolitan City, Nepal. NPRC Journal of Multidisciplinary Research, 2(10), 40-63. https://doi.org/10.3126/nprcjmr.v2i10.85865

Making Nepal's public information easier to find, understand and act on.

AI-powered aggregation | intelligent retrieval | cloud-ready delivery

Ashok Bhattarai
NP069811 | NP3F2509IT
Final Year Project | 2026